

Advanced Encryption Standard

AES-128 (Rijndael) Facts:

- Initiated in 1999 by NIST through a series of crypto conferences and published by NIST in 2001.
- “Rijndael” was the winner in a long list of ciphers including: MARS (IBM) and RC6 (RSA labs).
- AES-128 (Rijndael) is a 128-bit block cipher with variable length key (we’ll use a 128-bit key).
- More secure than triple-DES; designed to replace DES.
- Only public cipher approved by the NSA.
- 10 rounds of: **substitution, row shifting, column mixing, key adding.**
- Uses 8-bit binary in $GF(2^8)$ using $m(x) = x^8 + x^4 + x^3 + x + 1$; the first irreducible polynomial that the AES-128 developers found in their Finite Fields math textbook.
- No known practical attacks exist for a full 10 rounds of AES-128 (Rijndael).

Initialization: Each 128-bit string of data (block) is formatted into a 4x4 array of 2-hexit numbers:

111010101000.....000011000101

EA835CF00445332D655D98AD8596B0C5

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5

Here is one round of AES-128 (Rijndael):

S-box substitution (SubBytes):

The S-box table is used to map each 2-hexit element {x y} in the 4x4 array to another one located at row x and column y in the S-box:

87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6

The mapped (output) elements of the S-box are not a linear mathematical function of the input elements; they are derived using the multiplicative inverse of the input elements.

Row Permutation (ShiftRows)

Apply the following permutations to the elements in the rows:

Row 1: {1,2,3,4}

Row 2: {2,3,4,1}

Row 3: {3,4,1,2}

Row 4: {4,1,2,3}

87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6



87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

Additions and Multiplications in $GF(2^8)$ (MixColumns)

Number crunching... in $GF(2^8)$... with matrices... and binary bitwise XORs... and left bit shifts.

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} 87 & F2 & 4D & 97 \\ 6E & 4C & 90 & EC \\ 46 & E7 & 4A & C3 \\ A6 & 8C & D8 & 95 \end{bmatrix} = \begin{bmatrix} 47 & 40 & A3 & 4C \\ 37 & D4 & 70 & 9F \\ 94 & E4 & 3A & 42 \\ ED & A5 & A6 & BC \end{bmatrix}$$

Verify the first result: $\{02\} \times \{87\} + \{03\} \times \{6E\} + \{01\} \times \{46\} + \{01\} \times \{A6\} = \{47\}$

$$\{02\} \times \{87\} = (00001110) \oplus (00011011) = (00010101)$$

$$\{03\} \times \{6E\} = (11011100) \oplus (01101110) = (10110010)$$

$$\{01\} \times \{46\} = (01000110)$$

$$\{01\} \times \{A6\} = (10100110)$$

$$(00010101) \oplus (10110010) \oplus (01000110) \oplus (10100110)$$

$$= (10100111) \oplus (01000110) \oplus (10100110)$$

$$= (11100001) \oplus (10100110)$$

$$= (01000111)$$

$$= \{47\}$$

After a few rounds of MixColumns combined with Shiftrows, the output bits depend on ALL input bits.

Key Addition (AddRoundKey)

A new key is computed for EACH ROUND, based on the initial 128-bit key.

For example, if the first round key (derived from some initial 128-bit key) is:

AC7766F319FADC2128D12941575C006A

Then it is added by XOR to the 4x4 array from MixColumns:

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

 $\oplus$

AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

 $=$

EB	59	8B	1B
40	2E	A1	C3
F2	38	13	42
1E	84	E7	D6

Verify the first addition: $\{47\} + \{AC\} = (01000111) \oplus (10101100) = (11101011) = \{EB\}$

Therefore, the output ciphertext, after a single round is:

EB40F21E592E38848BA113E71BC342D6

Compare this to the initial input plaintext:

EA835CF00445332D655D98AD8596B0C5

AES decryption

Each step in a round of AES (SubBytes, ShiftRows, MixColumns, AddRoundKey) is invertible:

1. SubBytes is invertible using the **Inverse S-box** table.
2. Shiftrows is invertible by applying the inverse permutation for each row.
3. MixColumns is invertible by multiplying by the inverse of the encryption matrix:

inverse x encryption matrix = identity

$$\begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix} \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. AddRoundKey is invertible by adding the roundkey again.

Exercise:

1. AES MixColumns: calculate the elements “W”, “X”, “Y”, and “Z”.

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} 87 & F2 & 4D & 97 \\ 6E & 4C & 90 & EC \\ 46 & E7 & 4A & C3 \\ A6 & 8C & D8 & 95 \end{bmatrix} = \begin{bmatrix} 47 & 40 & A3 & W \\ 37 & X & 70 & 9F \\ 94 & E4 & Y & 42 \\ ED & A5 & A6 & Z \end{bmatrix}$$

Answers

1. "W" = 4C
"X" = D4
"Y" = 3A
"Z" = BC